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a $3x - 2y = 8$, $5x + 3y = 7$

Solution

Augmented matrix

$$\left[\begin{array}{cc|c} 3 & -2 & 8 \\ 5 & 3 & 7 \end{array} \right]$$

$$R_2 \rightarrow 3R_2 - 5R_1$$

$$\sim \left[\begin{array}{cc|c} 3 & -2 & 8 \\ 0 & 19 & -19 \end{array} \right]$$

$$\therefore 19y = -19$$

$$y = -1$$

Now

$$3x - 2(-1) = 8$$

$$\text{or, } 3x + 2 = 8$$

$$\text{or } x = \frac{6}{3}$$

$$\therefore x = 2$$

b $2x - y = 5, x - 2y = 1$

Soln

Augmented Matrix

$$\left[\begin{array}{cc|c} 2 & -1 & 5 \\ 1 & -2 & 1 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 2 & -1 & 5 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 3 & 3 \end{array} \right]$$

From the matrix

$$\begin{aligned} 3y &= 3 \\ \therefore y &= 1 \end{aligned}$$

substituting in $2x - y = 5$

$$\begin{aligned} 2x - 1 &= 5 \\ \text{or } x &= \frac{6}{2} \end{aligned}$$

$$\therefore x = 3 //$$

c. $x - 2y - 3z = -1$, $2x + y + z = 6$, $x + 3y - 2z = 13$
 Soln

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 2 & 1 & 1 & 6 \\ 1 & 3 & -2 & 13 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 0 & 5 & 7 & 8 \\ 0 & 5 & 1 & 14 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 0 & 5 & 7 & 8 \\ 0 & 0 & -6 & 6 \end{array} \right]$$

{ According to matrix

$$-6z = 6$$

$$\therefore z = -1$$

Substituting z , in eqⁿ from R_2

$$0x + 5y + 7(-1) = 8$$

$$\text{or, } 5y - 7 = 8$$

$$\text{or } y = \frac{15}{5}$$

$$\therefore y = 3$$

Finally, substituting both in eqⁿ from R₁

$$x - 2(3) - 3(-1) = -1$$

$$x - 6 + 3 = -1$$

$$x = -1 + 3$$

$$\therefore x = 2$$

d. $9y - 5x = 5, x + z = 1, z + 2y = 2$

Rearranging

$$-5x + 9y + 0z = 5, x + 0y + z = 1, 0x + 2y + z = 2$$

Augmented matrix

$$\sim \begin{bmatrix} -5 & 9 & 0 & : & 5 \\ 1 & 0 & 1 & : & 1 \\ 0 & 2 & 1 & : & 2 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 1 & : & 1 \\ -5 & 9 & 0 & : & 5 \\ 0 & 2 & 1 & : & 2 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & : & 1 \\ -5 & 9 & 0 & : & 5 \\ 0 & 2 & 1 & : & 2 \end{bmatrix}$$

$$R_2 \longrightarrow R_2 + 5R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & : & 1 \\ 0 & 9 & 5 & : & 8 \\ 0 & 2 & 1 & : & 2 \end{bmatrix}$$

$$R_3 \longrightarrow 9R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & 1 & : & 1 \\ 0 & 9 & 5 & : & 8 \\ 0 & 0 & -1 & : & 2 \end{bmatrix}$$

From R_3

$$-z = 2$$

$$\therefore z = -2$$

⑨

From R_2

$$9y + 5z = 8$$

$$\text{or } 9y - 10 = 8 \quad \left[\begin{array}{l} \text{From ⑨, } z = -2 \\ 5 \times -2 = -10 \end{array} \right]$$

$$\text{or } 9y = 18$$

$$\therefore y = 2$$

Similarly from R_1

$$x + z = 1$$

$$\text{or, } x + z = 1$$

$$\therefore x = 3$$

c $x + y + z = 6, x - y + z = 2, 2x + y - z = 1$
soln

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -6 & -18 \end{array} \right]$$

From the matrix

$$\begin{array}{l|l} -6z = -18 & -2y = -4 \\ z = 3 & y = 2 \end{array}$$

Now substituting y and z in eqⁿ from R₁,

$$x + y + z = 6$$

$$\text{or, } x + 2 + 3 = 6$$

$$\therefore x = 1$$

Q) $x - 2y - z = -7$, $2x + y + z = 0$, $3x - 5y + 8z = 13$

Solⁿ

Augmented matrix:

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 2 & 1 & 1 & 0 \\ 3 & -5 & 8 & 13 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 0 & 5 & 3 & 14 \\ 0 & 1 & 11 & 34 \end{array} \right] R_3 \rightarrow 5R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 0 & 5 & 3 & 14 \\ 0 & 0 & 52 & 156 \end{array} \right]$$

from R₃,

$$52z = 156$$

$$\therefore z = 3$$

Now,

from R₂

$$5y + 3(3) = 14$$

$$5y = 14 - 9$$

$$\therefore y = 1$$

Finally From R₁

$$x - (2)(1) - 1(3) = -7$$

$$\text{or } x = -7 + 2 + 3$$

$$\therefore x = -2 //$$

$$g. \quad 2x + 6y = -11, \quad 6x + 20y - 6z = -3, \quad 6y - 18z = -1$$

Soln

Augmented Matrix

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 6 & 20 & -6 & : & -3 \\ 0 & 6 & -18 & : & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 6 & -18 & : & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 0 & 0 & : & -91 \end{bmatrix}$$

$\therefore (0 = -91)$ is given by R_3
this system of linear equation has no solution.

b $2x + y + 4z = 1, x + y + 2z = 1, 4x + y + 10z = 1$
 3dⁿ

Augmented matrix

$$\sim \left[\begin{array}{ccc|c} 2 & 1 & 4 & 1 \\ 1 & 1 & 1 & 1 \\ 4 & 1 & 10 & 1 \end{array} \right]$$

$$R_1 \longleftrightarrow R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & 1 \\ 4 & 1 & 10 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 4R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & -3 & 6 & -3 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Since R_3 gives $0=0$, this has infinite solutions, hence let $z = k$

Now from R_2

$$-y + 2k = -1$$

$$\therefore y = 2k + 1$$

from R_1

$$1x + 1 + 2k + k = 1$$

$$\therefore x = -3k$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$R_1 - R_3 \rightarrow R_1, R_2 - R_3 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Since the rank of the coefficient matrix is 3 and the number of variables is 5, the system has infinitely many solutions.

Let $k = t$ and $l = s$

$$x = -3t$$

$$y = 1 - 3t - s$$

Q3. Test consistency and solve

a $x+y+z=6$, $x-y+z=8$, $5x-9y=-3$

solⁿ
In matrix form:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 5 & -9 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ -3 \end{bmatrix}$$

$$AX=b$$

For consistency

$$\rho(A) = \rho(A:b)$$

Now Augmented matrix

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & -1 & 1 & : & 8 \\ 5 & -9 & 0 & : & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 5R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 0 & : & 2 \\ 0 & -14 & -5 & : & -33 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 7R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 0 & : & 2 \\ 0 & 0 & -5 & : & -47 \end{bmatrix}$$

$$\rho(A) = 3, \rho(A:b) = 3$$

Now

from R_3 .

$$-5z = -47$$

$$z = \frac{47}{5}$$

Similarly

$$2y = 2$$

$$\therefore y = 1$$

$$x + (-1) + \frac{47}{5} = 6$$

$$\text{or } 5x - 5 + 47 = 30$$

$$\text{or, } 5x - 42 = 30$$

$$\text{or, } x = \frac{72}{5}$$

b $x+y+z = -2$, $2x+8y+5z = 5$, $x+2y-z = 2$
 Sol

Augmented matrix

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & -2 \\ 2 & 8 & 5 & 5 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

$R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & -2 \\ 0 & 6 & 3 & 9 \\ 0 & 1 & -2 & 4 \end{array} \right]$$

$R_3 \rightarrow 6R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -2 \\ 0 & 6 & 3 & 9 \\ 0 & 0 & -15 & 15 \end{array} \right]$$

$\rho(A) = \rho(A:b)$

Now,

from R_3

$$-15z = 15$$

$$z = -1$$

from R_2

$$6y + 3(-1) = 9$$

or $6y - 3 = 9$

$\therefore y = 2$

From
 R_1

$$x + (2) + (-1) = -2$$

$$\text{or } x + 1 = -2$$

$$\therefore x = -3$$

C $x + 2y - 3z = 9$, $2x - y + 2z = -8$, $3x - y - 4z = 3$

Sol

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 9 \\ 2 & -1 & 2 & -8 \\ 3 & -1 & -4 & 3 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 9 \\ 0 & -5 & 8 & -26 \\ 0 & -7 & 5 & -24 \end{array} \right]$$

$$R_3 \rightarrow 5R_3 - 7R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 9 \\ 0 & -5 & 8 & -26 \\ 0 & 0 & -31 & 62 \end{array} \right]$$

$$\rho(A) = \rho(A:b) = 3$$

From R_3 ,

$$-31z = 62$$

$$\therefore z = -2$$

From R_2

$$-5y + 8(-2) = -26$$

$$\text{or } 5y + 16 = 26$$

$$\text{or } 5y = 10$$

$$\therefore y = 2$$

From R_1

$$x + 2(2) - 3(-2) = 9$$

$$\text{or } x + 4 + 6 = 9$$

$$\text{or } x = 9 - 10$$

$$\therefore x = -1$$

$$Z = (A^{-1}B) = (A)^{-1}B$$

from R_3

from R_2

$$x = (P^{-1}) + 11$$

$$x = 514$$

$$x = 11$$

$$x = 5$$

e. $x - y + z = 1, x + z = -6, x + y - 2z = 3$

Augmented form

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 1 & 0 & 1 & -6 \\ 1 & 1 & -2 & 3 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -7 \\ 0 & 2 & -2 & 2 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -7 \\ 0 & 0 & -4 & 16 \end{array} \right]$$

$$\rho(A) = \rho(A:b) = 3$$

From R_3

$$\begin{aligned} -4z &= 16 \\ z &= -4 \end{aligned}$$

From R_2

$$\begin{aligned} y + (-4) &= -7 \\ \therefore y &= -3 \end{aligned}$$

from R_1

$$x - (-3) = 1$$
$$\text{or } x = -2$$

If the following system has non-trivial solution, find it

$$\begin{aligned} 2x_1 - 5x_2 + 8x_3 &= 0 \\ -2x_1 - 7x_2 + x_3 &= 0 \\ 4x_1 + 2x_2 + 7x_3 &= 0 \end{aligned}$$

Soln

Augmented matrix

$$\left[\begin{array}{ccc|c} 2 & -5 & 8 & 0 \\ -2 & -7 & 1 & 0 \\ 4 & 2 & 7 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 + R_1, R_3 \rightarrow R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 12 & -9 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2$$

$$\sim \left[\begin{array}{ccc|c} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\rho(A) < \text{no. of variables } (2 < 3)$

let,

$$x_3 = k$$

then,

From R_2

$$-12x_2 + 9k = 0$$

$$\therefore x_2 = \frac{3}{4}k$$

From R_1

$$2x_1 - 5 \times \frac{3}{4}k + 8k = 0$$

$$\text{or } 8x_1 - 15k + 32k = 0$$

$$\text{or } 8x_1 = -17k$$

$$\therefore x_1 = \frac{-17}{8}k$$